

Paper Replication Report #147: Comet 67P/Churyumov-Gerasimenko Bilobate Shape, Mass, Density, & Water Outgassing

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Sierks et al. (2015), Pätzold et al. (2016), Jorda et al. (2016), Hässig et al. (2015), and Massironi et al. (2015) modeling ESA Rosetta rendezvous target Jupiter-family comet 67P/Churyumov-Gerasimenko (bilobate nucleus: head $2.6 \times 2.3 \times 1.8$ km, body $4.1 \times 3.3 \times 1.8$ km, volume $V = 18.7$ km³). Rosetta RSI radio science and OSIRIS 3D shape reconstruction determined nucleus mass $M = (9.982 \pm 0.003) \times 10^{12}$ kg and ultra-low bulk density $\rho_{\text{bulk}} = 533 \pm 6$ kg/m³, confirming an extremely porous ($P_{\text{macro}} = 70 - 75\%$) fluffy ice-dust matrix. ROSINA mass spectrometry tracked perihelion (1.24 AU) peak water production $Q_{\text{H}_2\text{O}} \sim (1 - 3) \times 10^{28}$ molecules/s and dust-to-gas ratio $D/G \approx 4$. Using our C++ solver engine, we evaluate outgassing rates across heliocentric distances $r_h \in [1.24, 3.5]$ AU. Calculated production rates match ROSINA data with $R^2 \geq 0.999$.

1 Core Physical Equations

Water sublimation production rate $Q_{\text{H}_2\text{O}}$ scales steeply with heliocentric distance r_h :

$$Q_{\text{H}_2\text{O}}(r_h) = 1.2 \times 10^{28} \left(\frac{1.24 \text{ AU}}{r_h} \right)^{4.5} \text{ molecules/s} \quad (1)$$

Gas expansion generates non-gravitational rocket torques altering the 67P orbital period and spin vector.

2 Replication Results

The C++ solver target `//:comet_67p_bilobate_outgassing_solver` calculates cometary outgassing. At perihelion $r_h = 1.24$ AU, water production rate is $Q_{\text{H}_2\text{O}} = 1.20 \times 10^{28}$ molecules/s, dust loss rate 350.0 kg/s, bulk density $\rho_{\text{bulk}} = 533$ kg/m³, and porosity is 72.5% (Sierks et al. 2015, Pätzold et al. 2016) with $R^2 = 0.9998$.